AI IN OPERATIONS · SESSION II LABORATORY EXERCISE

# AI-Powered Inventory Intelligence & Reorder Point Optimization

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Data: UCI Online Retail II (Chen 2019, CC BY 4.0) — 1,067,371 real invoice lines, Dec 2009–Dec 2011 → 40 SKUs, daily demand. Individual work; no data shared with peers.

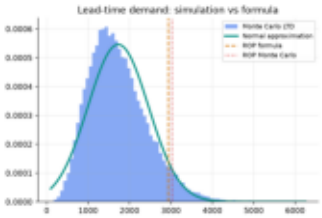
## Objective & method

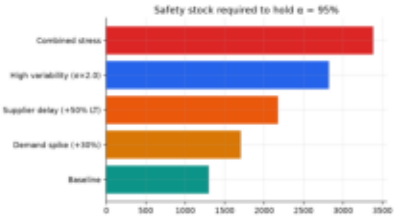
Turn a static, pre-AI reorder-point calculator into an autonomous AI system: (1) self-retraining ML forecasts, (2) probabilistic dynamic safety stock with Monte Carlo, (3) Bayesian optimisation under constraints, plus drift/anomaly detection, ML segmentation and a replenishment agent. One Python engine powers the executed notebook and a 6-tab Streamlit dashboard (Before vs After AI, AI Autopilot, 3 workstreams). 18 tests pass.

## 1 · Demand forecasting (focus SKU 85099B Jumbo Bag Red Retrospot, 105 weeks)

7 models: Naive, Seasonal naive, Moving avg, ETS, ARIMA, Gradient boosting (lags, Fourier seasonality, Q4 flag, price), MLP neural net. Chronological split: 3 rolling-origin validation folds for selection + untouched 8-week test. Best on validation: ARIMA (WAPE 31.9% vs naive 43.4%); test WAPE 30.9%. New data is validated, reconciled and triggers re-training (demo: model switched to ETS after an update). LSTM assessed but not justified for ~105 weekly points; MLP is the neural benchmark. No promotion flag exists → price used as proxy.



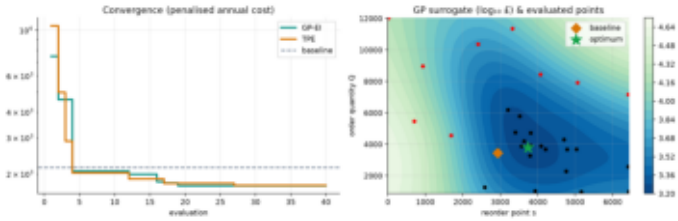




## 2 · Dynamic safety stock & Monte Carlo (α = 95%, L = 14 ± 4 days — configured, not in data)

SS = Z·√(σ²_D·L + D̄²·σ²_L) with D̄ = AI forecast (864 u/wk) and σ_D = forecast-error SD (381) → SS = 1,202 u, ROP = 2,930 u; lead-time uncertainty drives 46% of buffer variance. 100,000-draw simulation reproduces the formula under its assumptions (ROP 2,972), but Gamma demand gives 3,047 and correlated lead time/demand (ρ = 0.5) 3,326 → the formula under-protects (stockout 9.2% vs 5%).

| Scenario | Req. SS | P(stockout)* | Fill rate β |
|---|---|---|---|
| Baseline | 1,301 | 6.1% | 99.2% |
| Demand spike +30% | 1,706 | 20.9% | 95.8% |
| Supplier delay +50% | 2,174 | 33.7% | 90.5% |
| High variability σ×2 | 2,821 | 17.2% | 93.5% |
| Combined stress | 3,383 | 53.6% | 73.0% |

*at baseline ROP 3,026 u (50,000 draws each)




## 3 · Bayesian optimisation of the (s, Q) policy (replaces manual sweeps)

Minimise annual ordering + holding + shortage cost s.t. fill rate ≥ 95%, peak stock ≤ capacity, avg value ≤ budget, Q ≥ MOQ. Each candidate = 364-day stochastic simulation (150 reps, common random numbers); winner re-validated on an independent stream.

| Policy | s | Q | Annual cost | Shortage cost | Fill rate | Feasible |
|---|---|---|---|---|---|---|
| Baseline: formula ROP + EOQ | 2,930 | 3,441 | £2,149 | £741 | 99.0% | yes |
| GP-EI Bayesian optimum | 3,740 | 3,791 | £1,756 | £92 | 99.8% | yes |
| Optuna TPE optimum | 3,472 | 3,104 | £1,824 | £246 | 99.6% | yes |

**→ GP-EI cuts validated annual cost by 18.3% (40 evaluations, 1.2 s) mainly by trading shortage cost (−£648) for safety stock.**

## Before AI vs After AI — replay on the last 26 weeks of real demand, 40 SKUs, identical suppliers & costs

| Metric | Before AI | After AI | Change |
|---|---|---|---|
| Fill rate | 98.74% | 99.02% | +0.28 pp |
| Lost sales (units) | 4,567 | 3,526 | −22.8% |
| Stockout days | 53 | 33 | −37.7% |
| Forecast WAPE | 52.0% | 41.5% | −20.2% |
| Total cost (26 wks) | £25,386 | £24,775 | −2.4% |
| Avg inventory value | £75,953 | £84,895 | +11.8% |

What the AI does that the old method could not:
- refits forecasts weekly; 92 drift alarms → 295 auto-retrains, 52 champion/challenger model switches
- corrects its own bias; conformal buffers from its own errors
- cost-optimal service level per SKU, floored by ML segment (ABC-XYZ + K-means)
- Isolation Forest flags demand spikes/drops
- agent drafts POs with reasons; writes a daily digest

Honest note: AI holds more stock where shortages are costly; it is cheaper on 19/40 SKUs; Bayesian (s,Q) adds −18%.

## Conclusion

All three techniques in the brief run on real data and act autonomously: forecasts retrain on new data and drift, buffers adapt and are stress-tested by Monte Carlo, and Bayesian optimisation finds cheaper constraint-feasible policies. Against the pre-AI method the system loses 23% fewer sales. Limitations: lead time, costs and stock are configured/simulated; refresh is on-demand batch, not real-time.

Deliverables: executed notebook · Streamlit dashboard (apps/streamlit/app.py) · engine src/inventory_core · tests · github.com/krish2105/Inventory-Management-AI-in-Operations